PORTFOLIO OF SUBAYAN GHOSH

ABOUT ME

I am Subayan Ghosh, a highly motivated and results-driven B.Tech student in Electronics and Communication Engineering at the National Institute of Technology Durgapur, graduating in 2026 with a strong CGPA of 8.72/10. My academic journey has provided me with a robust foundation in core engineering principles, complemented by a deep dive into computer science fundamentals, including Data Structures, Algorithms, and Object-Oriented Programming. This interdisciplinary background has equipped me with a unique perspective, allowing me to approach complex problems with both hardware-level understanding and software-centric solutions.

My technical interests lie at the exciting intersection of Artificial Intelligence and Machine Learning, where I am passionate about transforming data into actionable insights and building intelligent systems that solve real-world challenges. I possess hands-on experience across the full stack of software development, with proficiency in Python, JavaScript, and C++, and expertise in frameworks like TensorFlow, React, Node.js, and Django. My project work, ranging from an AI-powered resume builder to a face-recognition attendance system, demonstrates my ability to conceptualize, develop, and deploy impactful applications.

My career aspiration is to evolve into a leading AI/ML Engineer, contributing to the development of cutting-edge intelligent systems and machine learning models. I am eager to leverage my analytical skills, programming prowess, and problem-solving mindset to design, implement, and optimize AI solutions that drive innovation and create significant value. I am particularly drawn to roles that involve deep learning, natural language processing, and computer vision, where I can apply my theoretical knowledge to practical, scalable applications.

What truly sets me apart is my blend of academic rigor, practical project experience, and a relentless drive for continuous learning and improvement. My leadership role in the Coding Club, success in competitive programming, and first-place win at the Smart India Hackathon underscore my ability to not only build but also innovate, lead, and collaborate effectively under pressure. I thrive on challenges, constantly seeking opportunities to expand my skill set and apply my knowledge to push the boundaries of what's possible with AI and machine Learning.

TECHNICAL EXPERTISE

My technical proficiency spans a wide array of programming languages, frameworks, tools, and core concepts, enabling me to tackle diverse engineering challenges and build robust, scalable solutions.

Programming Languages:

- Python:** My primary language for AI/ML development, data science, scripting, and backend web development (Django, Flask). Proficient in using libraries like TensorFlow, Pandas, NumPy, and Scikit-learn.
- C++:** Strong foundation in C++ for Data Structures and Algorithms, competitive programming, and performance-critical applications, demonstrating an understanding of

low-level memory management and efficient code.
• JavaScript:** Experienced in both frontend (React) and backend (Node.js, Express) development, enabling me to build full-stack web applications.
• SQL:** Proficient in querying and managing relational databases (MySQL, PostgreSQL) for data storage, retrieval, and manipulation.

Frameworks & Libraries:

• TensorFlow:** Hands-on experience in building, training, and deploying deep learning models for tasks such as image recognition, natural language processing, and predictive analytics.
• React:** Skilled in developing dynamic, responsive, and user-friendly single-page applications with modern UI/UX principles.
• Node.js & Express:** Proficient in building scalable and efficient backend RESTful APIs and server-side applications.
• Django:** Experienced in rapidly developing robust web applications with a focus on clean architecture and database integration.

Tools & Platforms:

• Git:** Expert in version control, collaborative development workflows, branching, merging, and pull requests.
• Docker:** Experienced in containerizing applications for consistent development, testing, and deployment across various environments, ensuring portability and scalability.
• Linux:** Proficient in command-line operations, scripting, and managing development/deployment environments on Linux-based systems.
• AWS:** Familiar with core Amazon Web Services (EC2, S3) for deploying and managing cloud infrastructure, particularly for web services and potential ML model deployment.
• Postman:** Regularly used for testing, documenting, and interacting with RESTful APIs during development.

Core Concepts & Methodologies:

• Data Structures & Algorithms:** Deep understanding and practical application of fundamental data structures (arrays, linked lists, trees, graphs, hash tables) and algorithms (sorting, searching, dynamic programming) for efficient problem-solving.
• Object-Oriented Programming (OOP):** Strong grasp of OOP principles (encapsulation, inheritance, polymorphism, abstraction) for designing modular, maintainable, and scalable software systems.
• REST APIs:** Expertise in designing, developing, and consuming RESTful web services for seamless communication between client and server applications.
• System Design:** Foundational understanding of architectural patterns, scalability considerations, database design, and distributed systems for building robust and high-performing applications.
• Machine Learning Fundamentals:** Solid understanding of supervised and unsupervised learning, model evaluation metrics, feature engineering, and neural network architectures.

PROJECT SHOWCASE

AI Resume Builder

Technol

Problem Statement:

The traditional resume creation process is often manual, time-consuming, and lacks optimization for Applicant Tracking Systems (ATS). Job seekers struggle to tailor their resumes effectively for specific roles, leading to lower interview rates despite relevant skills. This project addresses the need for an automated, intelligent solution to generate highly optimized and personalized resumes.

Solution Approach:

I developed a web-based AI Resume Builder leveraging Python, Streamlit for the interactive user interface, and the OpenAI API for advanced natural language processing. The core approach involved taking a user's raw resume content and a target job description, then using AI to analyze both, extract key skills and requirements, and intelligently rephrase or highlight relevant experiences to match the job's needs. The architecture facilitated secure API calls and real-time content generation.

Key Features:

- Job Description Parsing & Keyword Extraction:** Utilized OpenAI's language models to parse job descriptions, identify critical keywords, required skills, and responsibilities, ensuring the generated resume is highly relevant.
- Automated Content Generation & Optimization:** Dynamically generated resume sections (e.g., summary, experience bullets) by cross-referencing user input with extracted job requirements, optimizing phrasing for ATS compatibility.
- Customizable Templates & Formatting:** Provided various professional resume templates, allowing users to select and customize the layout while the AI focused on content optimization.
- Real-time Feedback & Suggestions:** Offered instant suggestions for improving resume sections based on AI analysis, guiding users to create more impactful descriptions.

Implementation Highlights:

- OpenAI API Integration & Prompt Engineering:** Successfully integrated the OpenAI API (GPT-3.5/GPT-4) for complex NLP tasks, requiring careful prompt engineering to achieve accurate keyword extraction, content generation, and contextual understanding.
- Streamlit for Interactive UI:** Built a responsive and intuitive user interface using Streamlit, allowing users to easily upload files, input job descriptions, and view generated resumes in real-time.
- Handling Large Text Data & API Rate Limits:** Implemented strategies for efficient processing of potentially large text inputs and managed API rate limits to ensure smooth user experience and prevent service interruptions.
- Security & Data Privacy:** Ensured user data was handled securely, with temporary storage and proper API key management.

Outcomes & Learnings:

The AI Resume Builder was successfully used by over 1000 students, significantly improving their resume quality and increasing their confidence in job applications. This project deepened my understanding of advanced NLP techniques, prompt engineering, and the challenges of building user-facing AI applications, particularly in managing API interactions and ensuring data privacy.

Smart Attendance System

Technol

Problem Statement:

Traditional attendance systems are manual, time-consuming, prone to human error, and susceptible to proxy attendance. Educational institutions and workplaces require a more efficient, accurate, and secure method for tracking attendance. This project aimed to develop an automated, face-recognition-based attendance system to address these inefficiencies.

Solution Approach:

I designed and implemented a full-stack smart attendance system. The backend, built with Node.js and Express, handled API requests, user authentication, and database interactions. The core face recognition logic was developed in Python, leveraging OpenCV and a pre-trained face-recognition library, processing video streams from webcams. MongoDB was used to store user profiles, facial encodings, and attendance records. The frontend, developed with React, provided a user-friendly interface for registration, attendance marking, and viewing reports.

Key Features:

- Real-time Face Detection & Recognition:** Utilized OpenCV for robust face detection in live video streams and a deep learning-based face recognition library to identify registered individuals with high accuracy.
- Automated Attendance Marking:** Automatically marked attendance for recognized individuals, logging the timestamp and user ID into the MongoDB database.
- User Registration & Profile Management:** Allowed administrators to register new users by capturing multiple facial images and storing their unique facial encodings securely.
- Interactive Dashboard & Reporting:** Provided a React-based dashboard for viewing daily, weekly, and monthly attendance reports, including late arrivals and absences, with filtering capabilities.

Implementation Highlights:

- Integrating Python Face Recognition with Node.js Backend:** Established a robust communication channel between the Python-based face recognition module (potentially via a microservice or child process) and the Node.js backend to handle real-time image processing and data transfer.
- Optimizing Face Recognition Performance:** Focused on optimizing the face recognition pipeline for speed and accuracy, experimenting with different models and pre-processing techniques to achieve a 95% accuracy rate in varying lighting conditions.
- Database Schema Design for Facial Encodings:** Designed a MongoDB schema to efficiently store and retrieve high-dimensional facial encodings, linking them to user profiles and attendance logs.
- Real-time Video Stream Handling:** Managed real-time video streams from webcams on the

frontend and efficiently transmitted frames to the backend for processing without significant latency.

Outcomes & Learnings:

The Smart Attendance System achieved a 95% accuracy rate, significantly reducing manual effort and improving the reliability of attendance tracking. This project provided invaluable experience in computer vision, deep learning for facial recognition, real-time system design, and the complexities of integrating different technology stacks (Python for ML, Node.js for backend, React for frontend).

E-commerce Web App

Technol

Problem Statement:

The demand for robust and scalable online shopping platforms continues to grow. Building a comprehensive e-commerce solution requires handling product management, user authentication, shopping cart functionality, secure payments, and an administrative interface. This project aimed to develop a full-stack e-commerce web application to demonstrate proficiency in modern web development practices.

Solution Approach:

I developed a complete e-commerce platform using the MERN stack. The frontend was built with React, utilizing Redux for state management, providing a dynamic and responsive user experience. The backend, powered by Node.js and Express.js, exposed a set of RESTful APIs for all core functionalities. MongoDB served as the NoSQL database for flexible data storage, and JSON Web Tokens (JWT) were implemented for secure user authentication. Payment processing was integrated using the Stripe API.

Key Features:

- User Authentication & Authorization:** Implemented secure user registration, login, and session management using JWT, with distinct roles for customers and administrators.
- Product Catalog & Management:** Developed features for browsing products by category, searching, viewing product details, and an admin interface for adding, updating, and deleting products.
- Shopping Cart & Order Processing:** Enabled users to add items to a persistent shopping cart, proceed to checkout, and manage their orders, including order history and status updates.
- Secure Payment Gateway Integration:** Integrated the Stripe API for secure and seamless credit card processing, handling payment intents and confirmations.
- Admin Dashboard:** A dedicated dashboard for administrators to manage products, users, orders, and view sales analytics.

Implementation Highlights:

- RESTful API Design & Development:** Designed and implemented a comprehensive set of RESTful APIs following best practices, ensuring clear endpoints, proper HTTP methods, and robust error handling.

• State Management with Redux:** Effectively managed complex application state on the frontend using Redux, ensuring data consistency and predictable state changes across components.
• Database Schema Design (MongoDB):** Designed a flexible and efficient NoSQL schema for MongoDB, accommodating product details, user information, order history, and reviews.
• Secure Authentication & Authorization:** Implemented industry-standard security practices for user authentication (hashing passwords, JWT) and authorization (middleware for role-based access control).
• Third-Party API Integration (Stripe):** Successfully integrated the Stripe API for payment processing, handling client-side and server-side interactions for secure transactions.

Outcomes & Learnings:

This project provided a deep, hands-on understanding of full-stack web development with the MERN stack, from database design and API development to frontend state management and secure payment integration. It solidified my skills in building scalable, secure, and user-friendly web applications, and exposed me to the complexities of managing a complete e-commerce lifecycle.

Weather Analytics Dashboard

Technol

Problem Statement:

Accessing and visualizing real-time weather data from various locations can be challenging without a centralized, interactive platform. Users often need to see current conditions, forecasts, and historical trends in an easily digestible format. This project aimed to create a web-based dashboard for real-time weather analytics.

Solution Approach:

I developed a web application using Python Flask for the backend, which served as an API proxy to fetch data from the OpenWeatherMap API. The frontend was built with standard HTML, CSS, and JavaScript, utilizing Chart.js for dynamic and interactive data visualization. The application allowed users to search for weather data by city and displayed various metrics in real-time charts.

Key Features:

• Real-time Weather Data Fetching:** Fetched current weather conditions and forecasts from the OpenWeatherMap API based on user-specified locations.
• Interactive Data Visualization:** Displayed key weather metrics (temperature, humidity, wind speed, pressure) using interactive line and bar charts powered by Chart.js.
• Location-Based Search:** Provided a search interface for users to input city names and retrieve relevant weather information.
• Responsive UI:** Designed the dashboard to be responsive, ensuring optimal viewing experience across various devices.

Implementation Highlights:

• API Integration & Data Handling:** Successfully integrated with the OpenWeatherMap API,

handling API keys, making HTTP requests, and parsing JSON responses to extract relevant weather data.
• Backend with Flask:** Developed a lightweight Flask backend to manage API calls, process data, and serve it to the frontend, acting as an intermediary to protect API keys and structure data.
• Frontend Data Visualization with Chart.js:** Implemented dynamic charts using Chart.js, updating them in real-time as new data was fetched, and configuring various chart types and options for clarity.
• Error Handling & User Feedback:** Implemented mechanisms to handle API errors (e.g., invalid city name, network issues) and provide clear feedback to the user.

Outcomes & Learnings:

This project enhanced my skills in API consumption, backend development with Flask, and frontend data visualization using JavaScript libraries. It provided practical experience in building a full-stack application that fetches, processes, and presents external data in an intuitive and interactive manner, reinforcing concepts of web development and data presentation.

PROFESSIONAL EXPERIENCE

Software Engineering Intern | ABC Technologies | May 2024 – Jul 2024

Overview:

During my internship at ABC Technologies, I was deeply involved in backend development, focusing on enhancing the performance and functionality of core services. My responsibilities included designing and implementing RESTful APIs, optimizing database interactions, and contributing to the deployment pipeline.

Key Achievements:

• Developed and maintained robust REST APIs using Django and PostgreSQL**, facilitating seamless data exchange for critical application modules, adhering to best practices for scalability and security.
• Reduced database query latency by 35%** across key application features by strategically implementing database indexing and introducing caching mechanisms (e.g., Redis), significantly improving application responsiveness.
• Streamlined service deployment by containerizing applications using Docker** and deploying them on AWS EC2 instances, ensuring consistent environments and reducing deployment time by 20%.
• Collaborated with a cross-functional team** to define API specifications and integrate backend services with frontend components, contributing to a cohesive product development cycle.

Technologies & Tools Used: Django, PostgreSQL, Docker, AWS EC2, REST APIs, Git, Python

Frontend Developer Intern | StartupHub | Dec 2023 – Feb 2024

Overview:

As a Frontend Developer Intern at StartupHub, I focused on building responsive and user-friendly interfaces for their web applications. My role involved translating UI/UX designs into high-quality code, optimizing for performance, and ensuring a seamless user experience.

Key Achievements:

- Built responsive and interactive user interfaces using React and Tailwind CSS**, delivering pixel-perfect implementations of design mockups across various screen sizes and devices.
- Improved page load speed by 40%** for critical user-facing pages through various optimization techniques, including code splitting, lazy loading components, and optimizing image assets, enhancing user engagement.
- Implemented reusable React components**, contributing to a modular and maintainable codebase that accelerated future feature development.
- Actively participated in daily stand-ups and code reviews**, providing constructive feedback and ensuring adherence to coding standards.

Technologies & Tools Used: React, Tailwind CSS, JavaScript, HTML, CSS, Git

EDUCATION & ACADEMIC BACKGROUND

- B.Tech in Electronics and Communication Engineering**

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CGPA: 8.72/10

Relevant Coursework:

- Data Structures and Algorithms
- Object-Oriented Programming
- Machine Learning
- Digital Signal Processing
- Microprocessors and Microcontrollers
- Database Management Systems
- Computer Networks

Academic Projects:

- Digital Image Processing Project:** Implemented various image processing algorithms (e.g., filtering, edge detection, segmentation) in Python, demonstrating a foundational understanding of image manipulation techniques relevant to computer vision.
- Embedded Systems Design:** Designed and prototyped a microcontroller-based system for environmental monitoring, integrating sensors and real-time data display, showcasing hardware-software co-design principles.

CERTIFICATIONS & CONTINUOUS LEARNING

My commitment to continuous learning is reflected in my pursuit of industry-recognized certifications and specialized courses, enhancing my expertise in cloud computing, machine learning, and data analytics.

AWS Cloud Practitioner | Amazon Web Services | 2024

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